

Retail Sales Analytics Report

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Tools Used:

Python
Pandas
Matplotlib
Power BI

Project Overview

The Retail Sales Analytics Project was developed to simulate a real-world retail business scenario where a Data Analyst is responsible for transforming raw sales data into meaningful business insights.

The project involved creating a realistic retail sales dataset containing over 1,000 transactions, performing comprehensive data cleaning, conducting exploratory data analysis, identifying business trends, and developing an interactive Power BI dashboard.

The primary objective of this project is to support business decision-making by analyzing sales performance, customer purchasing behavior, product performance, payment preferences, and geographic sales distribution.

Dataset Description

The dataset used in this project was self-created to simulate a realistic retail sales environment.

Dataset Information

- Number of Records: 1,225 (Raw Dataset)
- Dataset after Cleaning: 1,188 Records
- Industry: Retail Store Sales

Dataset Fields

- Transaction ID
- Customer ID
- Product Category
- Product Name
- Quantity
- Unit Price
- Total Amount
- Purchase Date
- Customer City
- Payment Method
- Customer Age
- Customer Gender

The dataset represents customer purchase transactions across multiple cities and product categories using various payment methods.

Data Cleaning Process

Several data quality issues were intentionally introduced into the raw dataset to simulate real-world business scenarios.

The following cleaning operations were performed:

- Handled missing values in Customer City, Payment Method, and Customer Age.
- Removed duplicate transaction records.
- Corrected inconsistent city names and product category names.
- Standardized payment method values.

- Converted Purchase Date into datetime format.
- Removed invalid values such as negative quantities, negative prices, and unrealistic customer ages.

The cleaned dataset was exported and used for further analysis.

Exploratory Data Analysis

The cleaned dataset was analyzed to identify business trends and customer purchasing behavior.

The following analyses were performed:

- Monthly Sales Trend
- Top Selling Products
- Revenue by Product Category
- Revenue by Customer City
- Customer Purchasing Patterns
- Payment Method Distribution
- Average Order Value
- Peak Sales Periods

Multiple visualizations were created using Python to better understand sales performance and identify business opportunities.

Business Insights

The analysis revealed several important business findings.

- Electronics products generated the highest overall revenue.
- Some product categories contributed significantly less revenue and require promotional strategies.
- Sales increased during peak shopping periods, indicating seasonal demand.
- Customers from major cities contributed the highest percentage of revenue.
- Multiple payment methods were actively used, with digital payments showing strong adoption.

These insights can help management improve inventory planning, marketing campaigns, and customer engagement strategies.

Dashboard Overview

An interactive dashboard was developed using Microsoft Power BI.

The dashboard includes:

- Total Revenue KPI
- Total Orders KPI
- Average Order Value
- Total Customers
- Total Quantity Sold
- Monthly Revenue Trend
- Revenue by Product Category
- Revenue by City (Map)
- Top Products
- Interactive Filters for Month, Category, City, Payment Method, and Year

The dashboard enables management to monitor business performance and make data-driven decisions.

Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase inventory levels for high-performing products.
- Launch promotional campaigns for underperforming categories.
- Focus marketing efforts on high-revenue cities.
- Expand digital payment options to improve customer convenience.
- Prepare inventory and staffing before peak sales seasons.
- Continue monitoring key performance indicators through the Power BI dashboard.

Conclusion

This project successfully demonstrated the complete retail analytics workflow, beginning with dataset creation and ending with an interactive business intelligence dashboard.

The combination of Python for data preparation and Microsoft Power BI for visualization enabled the extraction of valuable business insights that support informed decision-making.

This project reflects practical data analytics skills commonly used in real-world business environments.



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EXPLORER

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 - Data_cleaning
 - Data_cleaning.ipynb
 - Dataset
 - PowerBI_Dashboard
 - Reports
 - Business_Insights.md
 - Screenshots
 - Readme.MD
 - requirements.txt
- OUTLINE
- TIMELINE

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Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ... Python 3.1

Standardized text columns

```
[15] ✓ 0.1s
# Standardize city names by converting to title case
df["Customer_City"] = df["Customer_City"].str.title()

[16] ✓ 0.0s
# Remove leading/trailing whitespaces and convert product categories to title case
df["Product_Category"] = df["Product_Category"].str.strip().str.title()

[17] ✓ 0.0s
# Remove leading/trailing whitespaces and convert payment methods to title case
df["Payment_Method"] = df["Payment_Method"].str.strip().str.title()

[18] ✓ 0.0s
# Verify the unique values in the Customer_City column
df["Customer_City"].unique()

... array(['Karachi', 'Lahore', 'Islamabad', 'Multan', 'Peshawar',
        'Faisalabad', 'Unknown'], dtype=object)
```

Activate Windows
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```
[19] ✓ 0.0s

[20] ✓ 0.0s

... <class 'pandas.core.frame.DataFrame'>
Index: 1201 entries, 0 to 1202
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Transaction_ID         1201 non-null   object
1   Customer_ID            1201 non-null   object
2   Product_Category       1201 non-null   object
3   Product_Name           1201 non-null   object
4   Quantity               1201 non-null   int64
5   Unit_Price             1201 non-null   int64
6   Total_Amount           1201 non-null   int64
7   Purchase_Date          1191 non-null   datetime64[ns]
8   Customer_City          1201 non-null   object
9   Payment_Method         1201 non-null   object
10  Customer_Age           1201 non-null   float64
11  Customer_Gender        1201 non-null   object
dtypes: datetime64[ns](1), float64(1), int64(3), object(7)
memory usage: 122.0+ KB
```

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